ModelIt!

Teacher's Guide

Can a Robot Swim Like a Beaver?

How Nature Inspires Engineering Design

Quick Reference

Grade Level	Grade NE
NGSS Standard	4-LS1-1, 3-5-ETS1-1, 3-5-ETS1-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 4-LS1-1, 3-5-ETS1-1, 3-5-ETS1-3

Construct an argument that plants and animals have internal and external structures that function to support survival, growth, behavior, and reproduction. Define a simple design problem reflecting a need or want that includes criteria for success. Plan and carry out fair tests in which variables are controlled and failure points are considered.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how tail shape and body size affect swimming speed and energy use.
Disciplinary Core Idea	**LS1.A Structure and Function** — The beaver's flat tail is a structure adapted for the function of swimming — its shape pushes water efficiently.
Crosscutting Concept	**Structure and Function** — Students explore how the shape (structure) of a tail determines how well it works for swimming (function).

CAST Testing Alignment

MULTIPLE CHOICE:

An engineer is designing an underwater robot. She tests three tail shapes: flat, round, and pointed. The flat tail moves the robot fastest. Which best explains why? A) Flat is the lightest shape B) Flat pushes the most water backward C) Flat is the easiest to build D) Flat looks most like a fish

CONSTRUCTED RESPONSE:

A company wants to build a robot that swims like a beaver. Using what you learned from your model, explain what tail shape they should use and what tradeoff they need to think about with body size. Use the words 'speed' and 'energy' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Biomimicry — learning from nature to solve human engineering problems — is one of the fastest-growing fields in engineering. From bullet trains inspired by kingfisher beaks to swimsuits modeled on shark skin, nature's designs are helping engineers create better technology.

The Beaver's Tail

A beaver's tail is uniquely shaped — flat, wide, and paddle-like. It serves multiple functions: steering while swimming, slapping the water surface as a danger warning, propping the beaver up while gnawing trees, and storing fat for winter. The flat shape is ideal for pushing water, giving the beaver powerful swimming ability despite its heavy body.

What Is Biomimicry?

Biomimicry comes from the Greek words 'bios' (life) and 'mimesis' (to imitate). It is the practice of learning from nature's strategies to solve human design challenges. The idea is simple: nature has been solving engineering problems for 3.8 billion years through evolution. Why start from scratch when we can learn from what already works?

Famous Examples

Velcro was inspired by burrs that stuck to a dog's fur. Japanese bullet trains were redesigned with kingfisher-beak-shaped noses to reduce sonic booms. Wind turbines were improved by studying the bumpy edges of humpback whale fins. In every case, studying an animal adaptation led to better human technology.

Engineering Tradeoffs

Every design involves tradeoffs. A bigger engine is more powerful but heavier and uses more fuel. A bigger paddle pushes more water but takes more energy to move. Engineers must find the best balance — and nature has already found many of these balances through millions of years of natural selection.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify Tail Shape and Body Size as external components (design choices we make) and Swimming Speed and Energy Used as internal components (outcomes that change based on our choices).

E **Establish Relationships**
Students determine that flat tail shape increases swimming speed, larger body size decreases speed but increases energy use, and higher swimming speed also increases energy use.

V **Visualize & Model**
Students build a model showing how two design inputs produce two competing outputs — revealing the tradeoff.

E **Evaluate Outcomes**
Students run beaver tail vs. round tail scenarios, and big vs. small body scenarios, to see the tradeoff in action.

R **Revise & Extend**
Students add tail flexibility or water temperature to explore how additional factors change the design equation.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"What do a beaver and a robot have in common? More than you think! Today we discover how engineers learn from animals."*

Watch For

Show side-by-side images of animal features and technologies they inspired (Velcro/burrs, bullet train/kingfisher).

Slide 2: Learning Objectives

3 min

Say: *"Today we are going to figure out WHY a beaver's tail shape is so good for swimming — and use that knowledge to build something."*

Watch For

Have students read objectives. Pre-teach 'biomimicry' and 'tradeoff' with simple examples.

Slide 3: Big Question

3 min

Say: *"If you were building a robot that needed to swim, would you rather invent a brand new design or copy the best swimmer in nature?"*

Watch For

Have students turn and talk: What animal would you copy and why?

Slide 4: LEVER Framework

2 min

Say: *"Let us use our modeling process to test whether copying a beaver tail really works better."*

Watch For

Briefly review LEVER steps.

Slide 5: Activity 1: Components

8 min

Say: *"What design choices can we make? What outcomes will we measure?"*

Watch For

Guide sorting: Tail Shape and Body Size are choices WE make (external). Speed and Energy are results (internal).

Slide 6: Activity 2: Connections

8 min

Say: *"Here is the tricky part — making the tail flat makes speed go UP, but does energy go up or down too?"*

Watch For

Help students draw arrows. Key teaching moment: the TRADEOFF between speed and energy.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let us test: flat tail vs. round tail. Then big body vs. small body. Predict FIRST!"*

Watch For

Run three scenarios. Record predictions before each. Discuss surprises.

Slide 8: Discoveries

5 min

Say: *"Nature already figured this out! Beavers have flat tails because evolution tested every shape over millions of years."*

Watch For

Summarize findings. Show real biomimicry examples: shark skin swimsuits, whale fin turbines.

Slide 9: STEM Challenge

12 min

Say: *"Now use what the model taught you to design the best paddle. Your data should guide your choices!"*

Watch For

Distribute materials. Groups design, test, and iterate on paddle designs. Measure distance.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Tail Shape, Body Size
- **INTERNAL:** Swimming Speed, Energy Used

Relationships

- **Tail Shape to Swimming Speed:** VARIES — A flat tail (like a beaver's) pushes more water backward with each stroke, creating more forward thrust. A round tail pushes less water. A pointed tail pushes the least. Flat = fastest.
- **Body Size to Swimming Speed:** NEGATIVE (-) — A bigger body has more surface area dragging through water (water resistance). This slows it down. Smaller bodies move through water more easily.
- **Body Size to Energy Used:** POSITIVE (+) — A bigger body requires more energy to push through water. Like how it takes more effort to push a big box than a small one.
- **Swimming Speed to Energy Used:** POSITIVE (+) — Going faster always takes more energy — whether you are swimming, running, or driving. Speed costs energy.

Key Discoveries

- A flat tail (like a beaver's) pushes more water and creates more swimming speed
- A bigger body is harder to move through water — it uses more energy
- There is a TRADEOFF: being bigger gives more power but costs more energy
- Engineers study animal shapes because millions of years of evolution already solved these design problems

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Copying animals exactly will give the best design"	Biomimicry means being INSPIRED by nature, not making an exact copy. Engineers adapt natural designs for human needs. A robot does not need fur or teeth like a beaver — just the tail shape principle. Good biomimicry takes the IDEA and improves it for the specific use.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Bigger is always better in engineering"	Bigger often means more powerful, but it ALSO means more energy needed, more materials, and more cost. The best design is not always the biggest — it is the one that does the job with the least waste. This is why engineers care about EFFICIENCY, not just size.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Animals designed their own bodies"	Animals did not choose their features. Evolution through natural selection shaped them over millions of years. Beavers with slightly flatter tails swam better, survived longer, and had more babies. Over thousands of generations, the flat tail became standard. Nature does not design — it selects.